

▼ *Machine learning final project Report*

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▼ Team members

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Roles

Antonios: cleaning and splitting the data of the 1st problem to use it in the linear regression model and predicting recommendations in 3rd problem

Hussein: Applying the linear regression model and plotting it.

Hashem: cleaning the data of the 2nd problem and plotting the elbow diagram

Abdelrhman: Applying the k-means,PCA,T-SNE and plotting them.

=>We all worked on problem number three.

▼ process

Problem one process

first step : Checking the data and know its characteristics and know our target and predictor variables.

second step : Splitting the data to trainig and test data in order to use it in the linear regression model.

third step : Applying the linear regression model on the data.

fourth step: Plotting the results of the regression model.

fifth step: Evaluating our model by using the Mean square error method.

Problem two process

first step : Checking the data and ensuring that we will work only on numeric values.

second step : Using the elbow diagram to know the best NO. of clusters.

third step : Applying the k-means on the dataset then plotting it.

fourth step : Reducing the NO. of features using the PCA then applying the k-means.

fifth step : Reducing the NO. of features using the T-SNE then applying the k-means.

sixth step : Comparing the PCA and T-SNE

Problem three process

first step : loading data and checking them

second step: implementing collaborative filtering algorithm

third step: Normalizing the ratings and training the model

fourth step: predicting the recommendations

▼ References

▼ problem one References

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- [3] <https://koalatea.io/python-plot-kmeans-clusters/>
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- [3]<https://www.datacamp.com/tutorial/recommender-systems-python>
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